轉置矩陣(佔分 20 分)

問題描述：

若有 1 個 mxn 整數矩陣 a，則該整數矩陣 a 的轉置矩陣 a^T 表示如下圖。試撰寫程式能將整數矩陣 a 轉換成轉置矩陣 a^T。。

$$a=\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}，則 a^T=\begin{bmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{bmatrix}$$

輸入說明：

第 1 行輸入整數矩陣 a 的 m 列和 n 行，其中 $1\le\{m,n\}\le100$。

第 2 行輸入整數矩陣 a 內的元素 a_{11} a_{12} … a_{mn}，其中 $-999\le\{a_{11},a_{12},\ldots,a_{mn}\}\le999$。。

輸出說明：

輸出轉置矩陣 a^T。。

範例 1：

輸入：

2,2

1 2 3 4

輸出：

1 3 2 4

範例 2：

輸入：

1,5

-2 -4 -6 -8 -10

輸出：

-2 -4 -6 -8 -10

Transpose Matrix (20 points)

Problem Description:

Given an m x n integer matrix a, the transpose matrix aT of a is defined as transposing its rows and columns. Please write a program that converts the integer matrix a into its transpose matrix aT.

$$a=\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}，則 a^T=\begin{bmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{bmatrix}$$

Input Specification:

Line 1 contains the number of rows m and columns n of the integer matrix a, separated by a comma (e.g., m,n), where 1 <= m, n <= 100. Line 2 (and onwards) contains the elements a11, a12, ... amn of the integer matrix a, separated by spaces, where -999 <= elements <= 999.

Output Specification:

Output the transpose matrix aT. Elements in the same row should be separated by a single space.

Example 1:

```
Input:
2,2
1 2 3 4
Output:
1 3 2 4
```

Example 2:

```
Input:
1,5
-2 -4 -6 -8 -10
Output:
-2 -4 -6 -8 -10
```